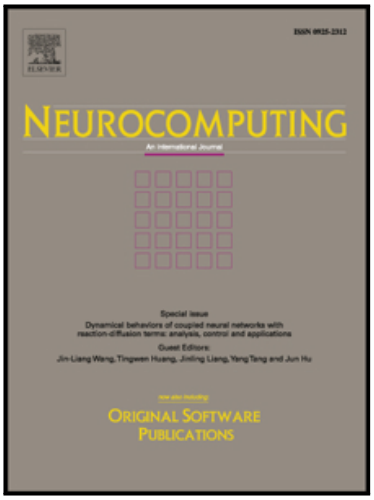
Communicated by Ass. Prof. L. Shao

Accepted Manuscript



Learning Part-based Mid-level Representation for Visual Recognition

Baodi Yuan, Jian Tu, Rui-Wei Zhao, Yingbin Zheng, Yu-Gang Jiang

PII: S0925-2312(17)31713-7
DOI: 10.1016/j.neucom.2017.10.062
Reference: NEUCOM 19042

To appear in: *Neurocomputing*

Received date: 15 March 2017
Revised date: 3 October 2017
Accepted date: 29 October 2017

Please cite this article as: Baodi Yuan, Jian Tu, Rui-Wei Zhao, Yingbin Zheng, Yu-Gang Jiang, Learning Part-based Mid-level Representation for Visual Recognition, *Neurocomputing* (2017), doi: 10.1016/j.neucom.2017.10.062

This is a PDF file of an unedited manuscript that has been accepted for publication. As a service to our customers we are providing this early version of the manuscript. The manuscript will undergo copyediting, typesetting, and review of the resulting proof before it is published in its final form. Please note that during the production process errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

ACCEPTED MANUSCRIPT

Learning Part-based Mid-level Representation for Visual Recognition

Baodi Yuan[a], Jian Tu[a], Rui-Wei Zhao[a], Yingbin Zheng[b], Yu-Gang Jiang[a]

[a] *School of Computer Science, Fudan University, Shanghai, China*
[b] *Shanghai Advanced Research Institute, Chinese Academy of Sciences, Shanghai, China*

Abstract

There exists a huge semantic gap between the low-level image representations and high-level semantics. To bridge such a gap, this paper proposes a mid-level image representation for visual recognition, where an image is represented based upon the response maps of local part filters. Each dimension of the mid-level representation indicates the likelihood of seeing a part in the input image. The part filters are trained using external data and need not to be fine-tuned on test data. To eliminate the possibly redundant similar parts occurring in different objects or scenes, we perform unsupervised clustering for part refinement. To alleviate the expensive computation of the response maps of the part filters, we further leverage sparse coding to accelerate the feature extraction process, which is ten times faster without significantly compromising the recognition accuracy. We evaluate the proposed mid-level representation on both image and video content recognition tasks and attain state-of-the-art results.

Keywords: Mid-level representation, part filter, scene recognition, event recognition, learning

Email addresses: 11110240022@fudan.edu.cn (Baodi Yuan), jtu13@fudan.edu.cn (Jian Tu), rwzhao14@fudan.edu.cn (Rui-Wei Zhao), zhengyb@sari.ac.cn (Yingbin Zheng), ygj@fudan.edu.cn (Yu-Gang Jiang)

ACCEPTED MANUSCRIPT

1. Introduction

Numerous efforts have been devoted to the construction of effective image representations. Based on vector quantization (*e.g.*, bag-of-words (BoWs) [1] and Fisher Vector [2]) of local descriptors like the SIFT [3], traditional approaches built up image-level representations for recognition and achieved great success. However, there always exists a huge semantic gap between the low-level image representations and high-level semantics. To bridge such a gap, mid-level visual representations are popularly constructed by linking each dimension with a semantic meaning. For example, when dealing with an image of polo sports, approaches using low-level representations are more interested in directly encoding visual words like edge, corner and color, while those using mid-level representations are engaged in discovering more meaningful parts, blocks or elements like helmet and horse head. Compared with low-level representations, mid-level representations have been demonstrated to be more natural and effective.

In this work, we propose a part-based mid-level visual representation, which is built upon the response maps of a large set of part filters. The part filters are learned on external training data with Deformable Part-based Models (DPMs) [4]. Following the classic DPM approach, we learn local part filters on publicly available image sets. The DPM approach was originally designed to represent objects. Pandey and Lazebnik [5] implemented a variant of DPM as scene representation and automatically discovered movable "regions of interest" (ROI's, or scene parts) by latent SVM training. This enables us to further extend the object part sets with scene parts. As scenes and objects have large differences in visual appearance, scene parts should supply complementary information to object parts.

Since different objects or scenes may share similar parts, we additionally conduct unsupervised clustering to discover and merge such common parts.

ACCEPTED MANUSCRIPT

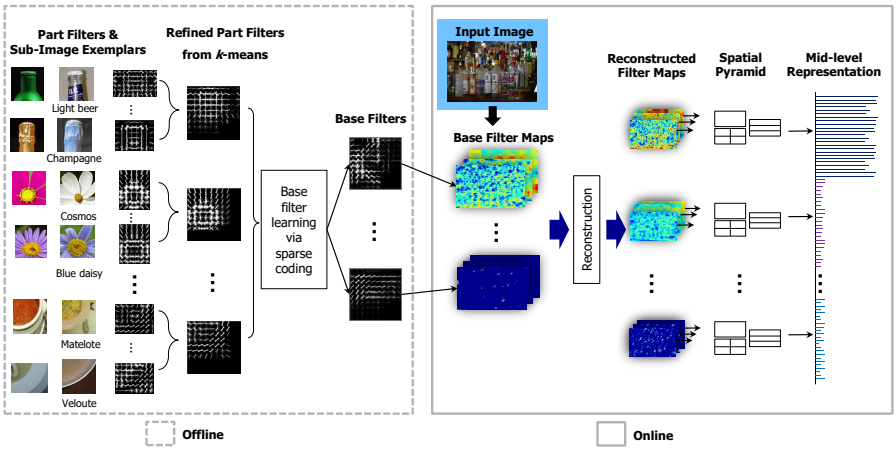


Figure 1: An illustration of our Part-based Mid-level Representation. We first learn a set of object (scene) part filters from existing object (scene) datasets (class names are shown below the part filters and sub-image exemplars). Next, we perform unsupervised clustering to merge similar part filters and produce refined part filters for the following sparse coding process. Base part filters are learned via sparse coding and then applied to an input image at multiple scales. The response maps of the refined part filters are reconstructed from those of the base part filters. Finally, we concatenate the max response values of each refined filter in each scale and spatial grid of a spatial pyramid of three levels, generating the final representation (#filters$\times$#scales$\times(1+2\times2+3\times1)$ dimensions). Note that base filter learning is offline.

Subsequently, given an input image, an exhaustive response computation can be performed for every part at every possible location within the image. Eventually, we concatenate the filter responses of the parts into a semantic level representation, with each dimension indicating the likelihood of seeing an part in the given image.

However, for filters based on features like the Histogram of Gradients (HOG), the calculation of the response maps is time consuming. Felzenszwalb *et al.* [4] proposed a cascade model to accelerate the response map calculation. For our representation, it becomes computationally infeasible to exhaustively calculate the filter responses as the number of parts grows. To address this problem and expedite the feature extraction process, we further introduce a sparse coding scheme. Starting from original parts, base parts are learned via sparse coding. The number of base parts is much smaller than that of the original ones. Hence,

ACCEPTED MANUSCRIPT

we only need to perform the exhaustive response map calculation for base parts, and use their responses to reconstruct the responses of the original parts. This speeds up the feature extraction process and therefore enables us to build a better representation that can capture more parts.

Figure 1 illustrates our proposed part-based mid-level representation. On the left side of Figure 1, the similar part filters are merged together via unsupervised clustering, *e.g.*, the center part from object classes "Cosmos" and "Blue daisy". This step reduces the redundancy in the part filters extracted from different DPMs. Sparse coding is then performed to speed up the calculation of the response maps of the refined filters.

We apply our Part-based Mid-level visual representation to image and video content recognition tasks. Our representation outperforms popular low-level and mid-level baseline representations on scene and event recognition benchmark datasets. Furthermore, our representation helps improve the state-of-the-art performance of Convolutional Neural Network (CNN) features [6][7][8] through fusing them together.

The rest of this paper is structured as follows. Section 2 introduces related works and Section 3 briefly reviews the deformable part-based models. Then we describe our mid-level visual representation in Section 4 and evaluate it on several benchmark datasets in Section 5. Finally, Section 6 draws our conclusions.

2. Related Works

Despite outstanding results of traditional image representations relying on encoding low-level features, the dictionaries learned from low-level features hardly have connection with category semantics. In order to solve this problem, mid-level representations were proposed, which have been successfully applied to

ACCEPTED MANUSCRIPT

various challenging tasks including object recognition, scene recognition, video action recognition, and so on.

Mid-level representations were originally proposed in object recognition. Farhadi *et al.* used attributes as mid-level semantics to describe objects and further learn new object categories with few or no examples [9]. The attributes proposed in [9] consist of shape, part and material and were further extended to include the outputs of localized object detectors in [10]. A similar framework which defines and selects attributes for zero-shot object learning was explored by Lampert *et al.* [11], Rohrbach *et al.* [12], and Yu *et al.* [13]. An iterative refinement procedure that jointly learns attributes and object classes was attempted by Wang and Forsyth [14] with multiple instance learning, and Wang and Mori [15] with an undirected graphical model. They showed that such a procedure leads to notable performance improvements. Torresani *et al.* selected 2,659 semantic concept classifiers (a semantic concept is similar to an attribute) and proposed to encode the outputs of classifiers as the Classemes descriptor for object recognition. Kuang *et al.* [16] tried to automatically discover parts with mid-level feature representation and select a subset of discriminant and non-redundant ones for object localization.

In addition to object recognition, mid-level representations were also explored in scene recognition. Vogel and Schiele [17] proposed to detect the occurrence of attributes locally over image regions and then represent an image with the frequencies of the detected local attributes. Instead of using binary occurrence labels, Parikh and Grauman [18] further proposed relative attributes to represent the relative strength of a local attribute. Singh *et al.* [19] proposed to learn discriminative patches under an unsupervised manner over every category of a testing dataset, upon which a mid-level visual representation was built. Several works [20, 21, 22] explored similar part-based pipelines to [19] and proposed

ACCEPTED MANUSCRIPT

different strategies to learn part detectors. Juneja *et al.* [20] proposed to learn part detectors with Exemplar SVMs and rank the distinctiveness of parts with entropy-rank curves. Doersch *et al.* [21] learned discriminative part detectors as attributes by extending the mean-shift algorithm to carry out density ratio estimation and heuristically ranking the importance of the extracted patches. Another work of Sun *et al.* [22] proposed D-Parts, which initialized part detectors by performing clustering over a large number of randomly cropped image parts and then learned the detectors via enforcing group sparsity.

The mid-level representations can easily be applied to other computer vision tasks. Hauptmann *et al.* [23] employed a set of semantic concept classifiers in video retrieval. Liu *et al.* [24] and Diba *et al.* [25] constructed mid-level representations for action recognition. Yuan *et al.* [26] extracted mid-level activity components hierarchically from videos for activity recognition. Xu *et al.* [27] built mid-level features with diffusion map to classify single-person actions and complex interactional activities. Mazloom *et al.* [28] performed feature selection per event on a set of concept detectors for video event classification. Berg *et al.* [29] proposed to automatically discover and model attributes by mining noisy Web image collections and applying those attributes for product image search. Sun *et al.* [22] also endeavored to apply their proposed representation to image co-segmentation tasks.

Compared with Object Bank [30][31] which trained hundreds of object detectors using off-the-shelf approaches like DPMs [4] and leveraged the image-level object classification outputs as features, our work is among part-based approaches, where the image representation is produced from the detection of local parts, not entire objects. Our approach is different from [10] because we learn parts automatically by building a DPM, while [10] used pre-defined and manually annotated parts. The outputs of object detectors were then con-

ACCEPTED MANUSCRIPT

solidated to generate the final image representation. As for other part-based mid-level approaches [19, 20, 21, 22], the key difference of our approach is that we use the same set of part detectors learned from external training data. These part filters are blind to testing datasets, which makes our representation less dataset-specific and more generic. This work is based on a previous conference paper [32] with many new contents. The extensions include exploring different types of part filters, employing clustering for redundancy reduction, performing sparse coding for feature extraction acceleration, and presenting new experiments on video content recognition.

3. Deformable Part-Based Models

To learn the basic object and scene part filters, we adopt the DPM framework proposed in [4] and its variant proposed in [5], which are briefly described in this section.

First, we compute the HOG features [33] to represent an image in a multi-scale feature pyramid. Using the multi-scale features is a standard setup for scale-invariant object detection. For the parameters in the HOG feature, *e.g.*, the numbers of pyramid levels and orientations, we follow the original settings described in [4][1] for learning object parts and those in [5][2] for learning scene parts.

A DPM represents a category by a mixture of star like models. A star model consists of a coarse root filter that roughly covers the entire object and several part filters that cover small parts. A filter (a rectangular template) defines a HOG window of a given size. Let $\mathbf{w}$ be a long vector obtained by concatenating the weight vectors of a filter in the row-major order. Let H be the HOG feature

[1] Codes from the authors of [4] are at `http://www.cs.brown.edu/~pff/latent/`.
[2] Codes from the authors of [5] are at `http://www.cs.unc.edu/~megha/DPM/`.

ACCEPTED MANUSCRIPT

pyramid and p specify the given location and scale in a given image. The filter response at p is given by the dot product between the filter vector $\mathbf{w}$ and $\phi(H, p)$, where $\phi(H, p)$ returns a vector concatenating the feature values in the corresponding subwindow of H with the top-left corner at p in the row-major order.

Denote by $z = (p_r, p_1, \ldots, p_n)$ a hypothesis of a category which specifies the positions of the root filter (p_r) and the part filters ($p_i, i = 1, \ldots, n$). The score of the category hypothesis z is given by

$$\begin{aligned} s(z) =& \mathbf{w}_r \cdot \phi(H, p_r) + \sum_{i=1}^{n} \mathbf{w}_i \cdot \phi(H, p_i) \\ & - \sum_{i=1}^{n} \mathbf{d}_i \cdot \phi_d(dx_i, dy_i) + b, \end{aligned} \tag{1}$$

where $\mathbf{w}_r$ represents the root filter vector, $\mathbf{w}_i$ represents the vector corresponding to the ith part filter, $\phi_d(dx_i, dy_i)$ computes the displacement of the ith part relative to its anchor position, $\mathbf{d}_i$ is a deformation cost vector for the ith part, and b is a bias term to make the output scores of multiple models comparable.

A latent SVM formulation is adopted to learn the DPM parameters which are denoted by a single vector $\beta = (\mathbf{w}_r, \mathbf{w}_1, \ldots, \mathbf{w}_n, \mathbf{d}_1, \ldots, \mathbf{d}_n, b)$. In the latent SVM, an example X is scored by a function in the following form:

$$f_\beta(X) = \max_z \beta \cdot \mathbf{\Phi}(X, z), \tag{2}$$

where z contains latent values (for a hypothesis), and $\mathbf{\Phi}(X, z)$ is a concatenation of the subwindows from a feature pyramid of X according to the latent values in z. Model training is then conducted by alternatively executing two steps. First, the parameters in β are fixed and optimization is performed for seeking the latent values over all the positive examples. After that, the latent values are

ACCEPTED MANUSCRIPT

fixed and the parameters in β are optimized using an objective function similar to a standard SVM.

The training procedure also includes several important settings such as the selection of suitable negative samples and the initialization of the model parameters. Unlike object datasets, scene datasets do not have object bounding box annotations. In the training phrase of the object DPM, the root filter weights are initialized by learning a standard linear SVM over the HOG features covering the annotated bounding box. For the scene DPM, the entire images from a positive set are used to initialize the root filters. We refer the readers to [4] and [5] for more details.

4. Mid-Level Representation with Part Filters

In this section, we describe the construction of the proposed mid-level representation with part filters. We first present an initialization to the part filters using object and scene training data, then the clustering process on the part filters, and finally a method that employs sparse coding for acceleration.

4.1. Object and Scene Categories

The initialization of a suitable set of part filters is important to the success of the mid-level representation. To collect a comprehensive set of part filters, as aforementioned, we explore not only objects but also scenes from publicly available large datasets.

For objects, we select object categories from the ImageNet [34]. ImageNet is an image database organized by the WordNet hierarchy [35]. Each class in ImageNet includes hundreds up to thousands of images to characterize a meaningful concept in the WordNet. Bounding box annotations for over 3,000

ACCEPTED MANUSCRIPT

popular object classes are available online[3]. The selected categories include not only traditional objects like "washer" and "banana", but also generalized ones like "volcano" and "bookshop", as suggested by [30]. We further refine this object category set by two criteria. First, only those with more than 100 example images are selected to make sure that the DPM has adequate training examples. Second, as the high-level classes in the WordNet hierarchy may be visually too diverse to train a good DPM, we only select object categories corresponding to the leaf nodes in the WordNet hierarchy. Finally, we select a fine set of 580 object categories, which cover a number of entities in the visual world, such as appliance (*e.g.*, "washer"), fruit (*e.g.*, "banana"), fabric (*e.g.*, "paper towel"), structure (*e.g.*, "bookshop"), *etc.*

For scenes, we select scene categories from the SUN397 dataset [36]. Scene categories in SUN397 correspond to the leaf nodes of an over-complete three-level hierarchy. We use the union of the official train/test split in [36] to train DPMs. The union includes 397 categories, where each category consists of 100 images. The selected subset covers a wide range of scenes from outdoor man-made scenes and natural scenes to indoor scenes.

We learn a DPM model for each individual object or scene class. An object DPM model is composed of one component which consists of one root filter and six part filters, as shown in Figure 2. For each scene category, we learn a more complex model containing two components following [5], where each component consists of one root filter and eight part filters, as shown in Figure 3. This produces 3,480 (580×6) object part filters and 6,352 ($397 \times 2 \times 8$) scene part filters.

[3]Download link: `http://www.image-net.org/download-bboxes`.

ACCEPTED MANUSCRIPT

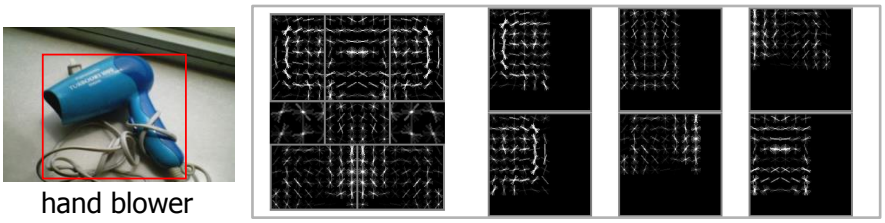


Figure 2: **Left:** example object category and an image from the category. **Middle:** deformable part-based model (DPM) of the category, which consists of a root filter and six part filters. **Right:** six part filters extracted from the DPM.

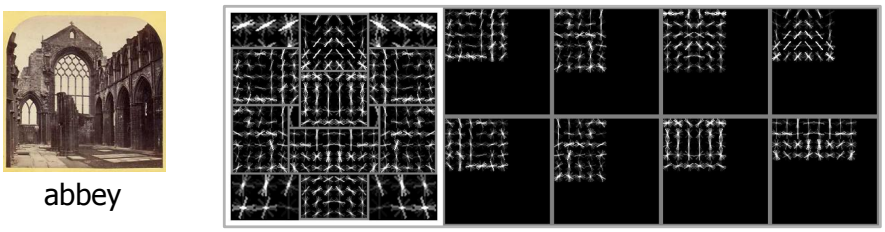


Figure 3: **Left:** example scene category and an image from the category. **Middle:** deformable part-based model (DPM) of the category, which consists of a root filter and eight part filters. **Right:** eight part filters extracted from the DPM.

4.2. Filter Refinement via Clustering

We can construct a mid-level representation of an image by directly applying the part filters to the image and afterwards pooling the corresponding response maps on a spatial pyramid. Nevertheless, different objects or different scenes may share similar parts, thus resulting in a significant amount of redundant information in the part filter set. To reduce the redundant information, we perform unsupervised K-means clustering and merge similar part filters.

The part filters extracted from DPMs are blocks of HOG cells of different sizes (*e.g.*, 6×8, 8×6 and 8×8). First, we align all part filters at the top-left corner and extend them to the same size with a zero-padding strategy. Let $S = \{\hat{\mathbf{p}}_k\}_{k=1}^{M}$ be a set of uniform part filters, where M is the total number of part filters. Performing unsupervised K-means clustering over the original part filters results in a smaller set of refined part filters $O = \{\mathbf{p}_k\}_{k=1}^{N} (N < M)$. The

ACCEPTED MANUSCRIPT

image region I in a cell of the spatial pyramid can be represented by a vector $\{e_k\}_{k=1}^N$ with each dimension derived by max pooling over the response map of a part detector $\mathbf{p}_k$, that is,

$$e_k = \max_{z_k \in \Omega_I} \mathbf{p}_k^T \phi(I, z_k), \tag{3}$$

where Ω_I denotes all possible image part locations.

As shown in Figure 1, similar parts are merged into a single part. The K-means clustering process makes the representation more semantically balanced, since some frequently "repetitive" parts in the original part filter set (*e.g.*, the center parts of similar flowers) may dominate the infrequent ones.

4.3. Acceleration via Sparse Coding

It is computationally expensive to exhaustively calculate the local responses at all possible locations. To address this issue, we propose a sparse coding based method to accelerate the intensive feature extraction process.

Denote the dimension of a part filter as L, so $\mathbf{p}_k \in \mathbb{R}^{L\times 1}$. $\mathbf{P} = (\mathbf{p}_1, \mathbf{p}_2, \cdots, \mathbf{p}_N)$ is an array of all the refined part filters, where $\mathbf{P} \in \mathbb{R}^{L\times N}$. The response of the part filter at an arbitrary location z is $\mathbf{P}^T\phi(I, z)$. The response maps of all the part filters can thus be calculated by searching all possible locations, leading to

$$\mathbf{R} = \{\mathbf{P}^T\phi(I, z) | z \in \Omega_I\}. \tag{4}$$

We use the open source SPArse Modeling Software[4] provided by the authors of [37] and [38] to perform sparse coding over the part filter array $\mathbf{P}$. Dictionary learning along with sparse coding is performed by minimizing the following

[4]Available at `http://www.di.ens.fr/willow/SPAMS/`.

ACCEPTED MANUSCRIPT

objective function:

$$L = \min_{\mathbf{D} \in C} \frac{1}{n} \sum_{k=1}^{N} \frac{1}{2} ||\mathbf{p}_k - \mathbf{D}\alpha_k||_2^2, \qquad \text{s.t.} \quad ||\alpha_k||_0 \leq \lambda, \tag{5}$$

in which C is a constraint set for the dictionary and λ is the cardinality to control the sparsity level of sparse codes.

The part filter array $\mathbf{P}$ can be decomposed into the product between a dictionary $\mathbf{D}$ and a series of sparse codes $A = (\alpha_1, \alpha_2, \cdots, \alpha_N)$:

$$\mathbf{P} \approx \mathbf{D}A, \tag{6}$$

where $\mathbf{D} \in \mathbb{R}^{L \times K}$, $A \in \mathbb{R}^{K \times N}$, and K is a user-determined parameter in the sparse coding algorithm.

By representing $\mathbf{D} = (\mathbf{w}_1, \mathbf{w}_2, \cdots, \mathbf{w}_K)$ as an array of base part filters $\{\mathbf{w}_k\}_{k=1}^{k=K} \in \mathbb{R}^{L \times 1}$ and applying Eq. (4) to Eq. (6), we obtain the following approximation to the response maps:

$$\begin{aligned} \mathbf{R} &\approx \{A^T \mathbf{D}^T \phi(I, z) | z \in \Omega_I\} \\ &\approx A^T \{\mathbf{D}^T \phi(I, z) | z \in \Omega_I\} \\ &\approx A^T \mathbf{R}', \end{aligned} \tag{7}$$

where $\mathbf{R}' = \{\mathbf{D}^T \phi(I, z) | z \in \Omega_I\}$ includes the response maps of the base filters $\{\mathbf{w}_k\}_{k=1}^{k=K}$ over the whole image region.

Figure 4 shows 100 base filters learned through dictionary learning and sparse coding over the refined object filters (by setting $K = 100$). It is interesting to observe that some basic shapes can be found from the base filters.

The response map calculation of all the part filters can then be replaced by two steps. First, calculate the response maps of the base part filters. Second, multiply the above response maps, *i.e.*, $\mathbf{R}'$, by a sparse matrix A^T to reconstruct

ACCEPTED MANUSCRIPT

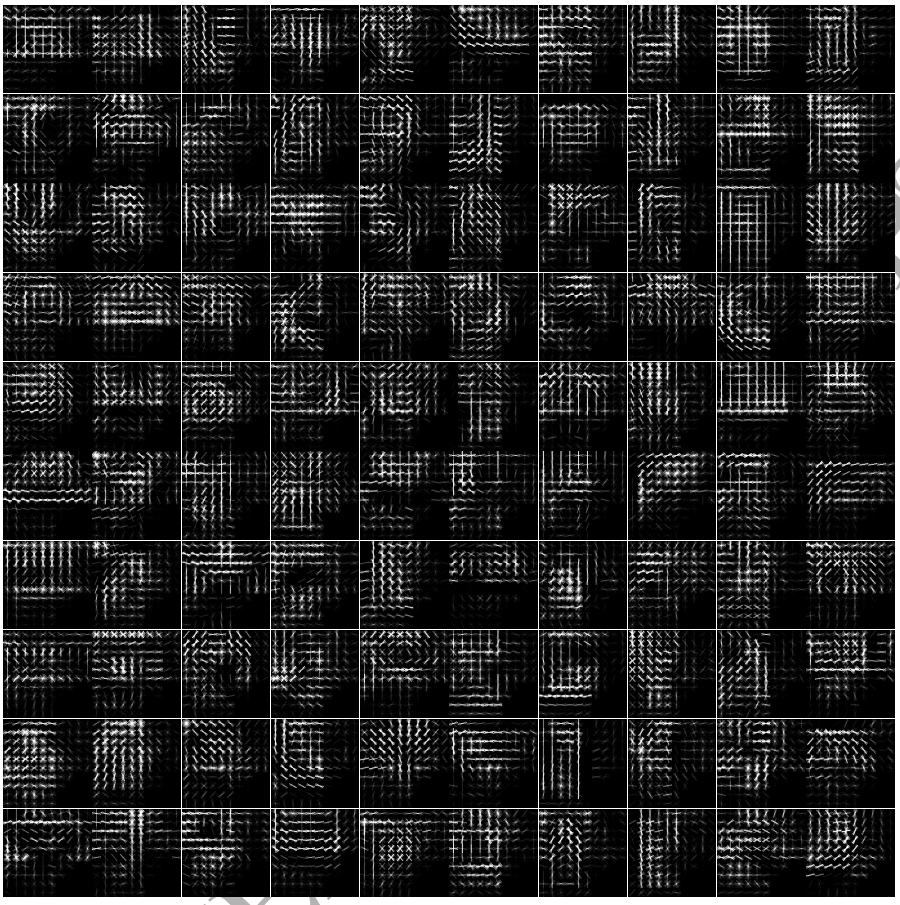


Figure 4: Base part filters learned via sparse coding. Each base filter is a HOG block of 10×10 cells. Note that the original part filters before applying sparse coding are aligned at top-left corner and padded with empty cells below and on the right side. It is interesting to observe that some basic shapes can be found on the learned base filters.

the response maps of the original part filters, *i.e.*, $\mathbf{R}$. Since the number of the base part filters is much smaller than that of the original part filters in the first step and the coefficient matrix A^T is sparse in the second step, the response map approximation via sparse coding dramatically accelerates the feature extraction process.

ACCEPTED MANUSCRIPT

ACCEPTED MANUSCRIPT

5. Experiments

In this section, we evaluate the proposed representation on both image and video content recognition tasks. We first introduce the experimental setup, followed by results and discussions.

5.1. Datasets

To show the effectiveness of our proposed mid-level visual representation, we evaluate the method on four popular datasets for image (MIT Indoor [39], Fifteen Scenes [40] and UIUC Sports [41]) and video (CCV [42]) content recognition.

MIT Indoor [39]. It contains 15,620 images labeled over 67 classes. Each class contains at least 100 images. Classes in the dataset are grouped into 5 categories, *i.e.*, store (*e.g.*, bakery and grocery), home (*e.g.*, bedroom and nursery), public spaces (*e.g.*, library and cloister), leisure (*e.g.*, gym and bar) and working place (*e.g.*, office and warehouse). This is generally more difficult than outdoor scenes, because some indoor scenes are very similar and discriminative features are needed to separate them. We use the official train/test split defined in [39], which uses 80 images per class for training and 20 for testing.

Fifteen Scenes [40]. This dataset includes a total of 15 indoor (*e.g.*, bedroom and kitchen) and outdoor (*e.g.*, forest and mountain) scene classes. The number of images in each class ranges from 210 to 410. Following [41], we randomly select 100 images from each class for training and use the rest for testing. We repeat the random selection process for ten times and present the mean performance over the ten runs.

UIUC Sports [41]. This dataset contains 8 sports event classes (rowing, badminton, polo, bocce, snowboarding, croquet, sailing and rock climbing) with 137-250 images per class. Following [41], for each category, we use 70 randomly

ACCEPTED MANUSCRIPT

images for training and 60 for testing. Like the Fifteen Scene dataset, we also report the mean performance over ten runs on this dataset.

Columbia Consumer Videos (CCV) [42]. This is a collection of 9,317 unedited consumer videos from YouTube labeled over 20 categories (*e.g.*, biking, soccer and birthday). The average length of videos in CCV is around 80 seconds. CCV is more difficult than the other datasets, as some classes are fairly complex and correlated (*e.g.*, wedding ceremony, wedding reception and wedding dance). We adopt the official train/test split defined in [42] in our experiments.

For the first three datasets, we adopt classification accuracy as the main performance measure. As some existing works reported average precision (AP) on these datasets, we also adopt AP and report mean AP (mAP) as the overall performance measure over all the classes. For CCV, we only report mAP as it is the only official measure on this dataset.

5.2. Implementation Details

The process of part filter learning generates 3,480 object part filters and 6,352 scene part filters, which are then uniformly extended to the same size ($10 \times 10 \times 31$) by the zero-padding strategy. Based on the original filters, the clustering process produces 2,000 and 4,000 refined filters respectively by merging relatively similar ones. After that the base filters are learned from the refined filter set via sparse coding. The number of the base filters and the sparsity ratio are set as 100 and 15% respectively. Key parameters of the clustering will be evaluated later.

At the feature pooling phase, we concatenate the max response values of each part filter in each image scale and spatial grid to generate the final representation ($\#\text{filters} \times \#\text{scales} \times (1 + 2 \times 2 + 3 \times 1)$ dimensions). For the video dataset CCV, we extract a frame from each 2s clip and perform max-pooling over the representations of all the frames to generate the final video-level representation.

ACCEPTED MANUSCRIPT

To alleviate the effect of insufficient training samples, data augmentation is adopted to generate two representations for each sample (original images and their left-right mirrors). Following existing works, for the CCV dataset, we train one-vs-all χ^2 kernel SVMs, and for all the other datasets, we use one-vs-all linear SVMs following [32]. The parameters of the SVM classifiers are estimated by cross validation.

5.3. Results and Discussions

In this subsection, we discuss the results of our part-based mid-level representation, named as PMR. For the three image datasets, we compare with two well-known low-level image representations GIST-color [43] and Spatial Pyramid (SP) [40] and three mid-level image representations Object bank (OB) [30], Classemes [44] and Discriminative Part Detectors (D-parts) [22]. For the CCV dataset, we compare with several baselines presented in [45] and two mid-level image representations: Object Bank and Classemes.

MIT Indoor. We first report and analyze the results on MIT Indoor, which are summarized in Table 1. From the table we can see that all the mid-level representations, including our PMR, outperform the two low-level representations. This validates the advantages of the mid-level representations in bridging the semantic gap.

Compared with Object Bank and Classemes, the performance of our PMR is significantly better in terms of both accuracy and mAP. We attribute this to the fact that the PMR is built on parts that carry valuable fine-grained clues, while Object Bank is based on the entire object level detections and Classemes contains a collection of object and scene recognition results also on the entire image-level. Compared with D-parts [22] which achieves an mAP of 51.4% by learning 4,926 part filters (detectors) from the 67 classes in MIT Indoor, our PMR with only 2,000 object part filters produces 54.2%. Notice that D-parts

ACCEPTED MANUSCRIPT

Table 1: Performance on the MIT Indoor dataset, measured by both classification accuracy and mAP. We compare our part-based mid-level representation (PMR) with two low-level baselines (GIST and SP) and three existing mid-level representations. Results without a source reference are based on our implementation.

Approach	ACC	mAP	Source
GIST-color [43]	29.7%	-	[5]
SP [40]	34.4%	-	[5]
Classemes [44]	42.8%	45.5%	-
Object Bank [30]	37.6%	-	[30]
D-Parts [22]	-	51.4%	[22]
PMR (object only)	52.8%	54.2%	-
PMR (scene only)	51.8%	53.0%	-
PMR (all)	53.8%	55.2%	-

adopted complex learning methods to generate the representation based on parts learned from the target dataset, while our PMR is based on external data that is independent from the target testing datasets. Learning from external data may bring in valuable information, which might be a reason that PMR is better. Another advantage of doing so is that PMR can be deployed on any new test data without the need of re-learning the part filters like D-parts.

We also study the contributions of the object and the scene part filters separately. As shown in Table 1, the object part filters demonstrate slightly better results, although the number of object filters is only half of that of the scene filters. Possible reasons are two-fold. First, objects are probably more valuable clues in inferring visual semantics, as for many classes objects are the key subjects and scenes are the backgrounds. Second, the object part filters are learned on images with precise spatial bounding boxes, but we do not have such spatially annotated data to learn the scene part filters.

The per-class AP results of the 67 classes are listed in Table 2. The difference between results using object and scene filters is reported in the "O-S" columns. While the performance of the two sets of filters is very close for most classes, the gap reaches up to 13% for some classes. For example, the object filters

ACCEPTED MANUSCRIPT

Table 2: Per-class AP (%) on the MIT Indoor dataset, using "PMR (object only)" (O), "PMR (scene only)" (S) and "PMR (all)" (A). The classes are sorted in descending order of the "PMR (all)" performance. "O-S" indicates the difference between "O" and "S".

	O	S	A	O-S		O	S	A	O-S
cloister	90	96	95	-6	clothingstore	53	52	53	1
elevator	91	90	92	1	hairsalon	49	54	53	-5
greenhouse	93	90	92	3	kitchen	54	51	53	3
corridor	87	85	87	2	poolinside	49	50	52	-1
pantry	87	80	87	7	airport_inside	44	51	49	-7
church_inside	82	83	84	-1	grocerystore	48	44	49	4
inside_bus	84	83	84	1	computerroom	49	40	48	9
bowling	78	79	79	-1	dentaloffice	47	42	48	5
buffet	78	75	76	3	fastfood_restaurant	45	38	47	7
studiomusic	77	64	76	13	warehouse	47	44	47	3
florist	69	77	74	-8	restaurant_kitchen	39	47	46	-8
laundromat	74	73	74	1	children_room	43	44	45	-1
stairscase	73	74	74	-1	bookstore	41	40	44	1
inside_subway	72	74	73	-2	locker_room	43	38	42	5
closet	73	70	72	3	bathroom	41	40	41	1
trainstation	72	70	72	2	dining_room	40	39	40	1
classroom	71	66	70	5	gym	41	36	40	5
casino	60	70	69	-10	restaurant	37	34	40	3
auditorium	67	63	68	4	lobby	37	32	35	5
nursery	69	65	68	4	museum	37	31	35	6
concert_hall	70	60	66	10	laboratorywet	29	38	34	-9
meeting_room	69	56	66	13	livingroom	30	29	34	1
movietheater	59	70	66	-11	mall	34	34	34	0
prisoncell	64	66	66	-2	artstudio	30	30	31	0
bedroom	65	63	65	2	operating_room	30	28	31	2
kindergarden	61	61	64	0	bakery	31	25	28	6
garage	62	59	63	3	jewelleryshop	27	25	28	2
hospitalroom	59	61	63	-2	shoeshop	29	25	28	4
subway	64	57	62	7	bar	27	26	27	1
tv_studio	57	61	60	-4	deli	19	28	27	-9
library	56	57	58	-1	waitingroom	27	26	26	1
gameroom	51	57	55	-6	toystore	24	24	23	0
winecellar	58	50	55	8	office	11	9	10	2
videostore	55	48	54	7					

perform significantly better on "studio music", "concert_hall", "meeting_room" and "computer room", while the scene filters perform better on "movie theater", "casino", "deli", etc.

Combining the object and the scene part filters can further improve the results. We generate two representations based on the two kinds of filters sepa-

ACCEPTED MANUSCRIPT

Table 3: Performance on the Fifteen Scenes dataset, measured by both classification accuracy and mAP. Results without a source reference are based on our implementation.

Approach	ACC	mAP	Source
GIST-scolor [43]	68.0%	71.9%	-
SP [40]	76.9%	83.0%	-
Classemes [44]	82.7%	87.6%	-
Object Bank [30]	80.9%	-	[30]
D-Parts [22]	-	86.0%	[22]
PMR (object only)	87.4%	92.5%	-
PMR (scene only)	86.9%	92.0%	-
PMR (all)	87.5%	92.7%	-

rately, and combine the prediction results using simple average fusion. Overall, as shown in Table 1, the results can be improved to 53.8% in accuracy and 55.2% in mAP, which are significantly better than all the compared alternative methods.

Fifteen Scenes & UIUC Sports. We summarize the results on Fifteen Scenes in Table 3 and UIUC Sports in Table 4. As the results demonstrate almost the same trend on the two datasets, we discuss them together. In addition to Classemes, we also implement GIST-color and SP on both datasets for comparison. For GIST-color, we use the software released by the authors of [43][5]. For SP, we use a codebook of 300 codewords and 3 spatial pyramid levels. As shown in Table 3 and Table 4, our mid-level representations again outperform all the alternative methods. We also observe that, on both datasets, the gain from fusing object and scene part filters is very limited. One possible reason is that Fifteen Scenes and UIUC Sports are easier with much fewer classes, and the performance is already saturated with only one type of the part filters.

CCV. Table 5 compares our PMR representation with several baseline low-level features and alternative mid-level representations on the CCV dataset. The results of Object Bank and Classemes are obtained based on the codes

[5]Available at `http://people.csail.mit.edu/torralba/code/spatialenvelope/`.

ACCEPTED MANUSCRIPT

Table 4: Performance on the UIUC Sports dataset, measured by both classification accuracy and mAP. Results without a source reference are based on our implementation.

Approach	ACC	mAP	Source
GIST-color [43]	72.6%	79.4%	-
SP [40]	81.4%	89.3%	-
Classemes [44]	84.9%	91.6%	-
Object Bank [30]	76.3%	-	[30]
D-Parts [22]	-	86.4%	[22]
PMR (object only)	87.3%	92.9%	-
PMR (scene only)	87.7%	92.6%	-
PMR (all)	88.1%	93.2%	-

Table 5: Performance on the CCV dataset. Results without a source reference are based on our implementation.

Approach	mAP	Source
GIST	32.5%	[45]
STIP	44.9%	[45]
SIFT	52.3%	[45]
Object Bank [30]	38.4%	-
Classemes [44]	44.8%	-
PMR (object only)	55.1%	-
PMR (scene only)	54.8%	-
PMR (all)	60.1%	-

from the original authors. For all the methods, video-level representation is generated by performing max-pooling on the representations of frames, which are sampled with 2s intervals.

As can be seen, the mAP of our method is more than 10% higher than the alternative mid-level representations. It also significantly outperforms the popular low-level features on this dataset like the computationally expensive spatial temporal interest points (STIP) and the SIFT. By combining "PMR (object only)" and "PMR (scene only)", the mAP number further increases 5% to 60.1%. As CCV is more complex than the three image datasets, the gain from fusing the object and the scene part filters indicates that the two kinds of part-level filters are fairly complementary under complex situations (in contrast to the Fifteen Scenes and UIUC Sports where the performance is saturated). In

ACCEPTED MANUSCRIPT

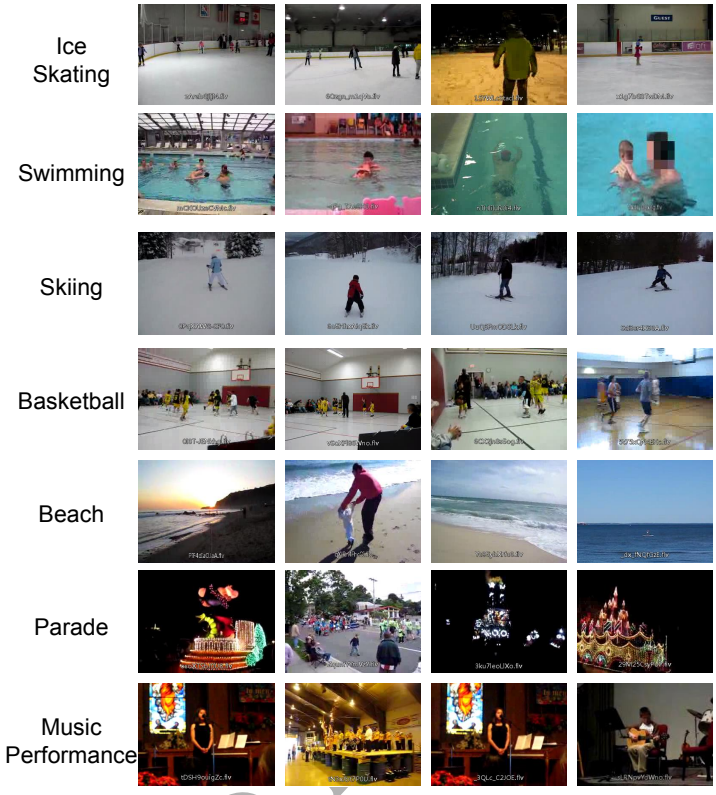


Figure 5: Example frames of several CCV categories. For each category, we show a frame from each of the top four videos in the test set recognized by the "PMR (all)" approach.

other words, high-level categories like "parade" require clues from both objects and scenes to be accurately identified. Figure 5 shows example frames of several CCV categories.

5.4. Effect of Clustering and Sparse Coding

As elaborated earlier, our PMR involves a clustering process to remove redundant information and a sparse coding based process for acceleration. In this subsection, we study the effect of both processes, using the MIT Indoor dataset.

Figure 6 shows the performance with different numbers of clusters. Recall that before clustering there are 3,480 object part filters and 6,352 scene part fil-

ACCEPTED MANUSCRIPT

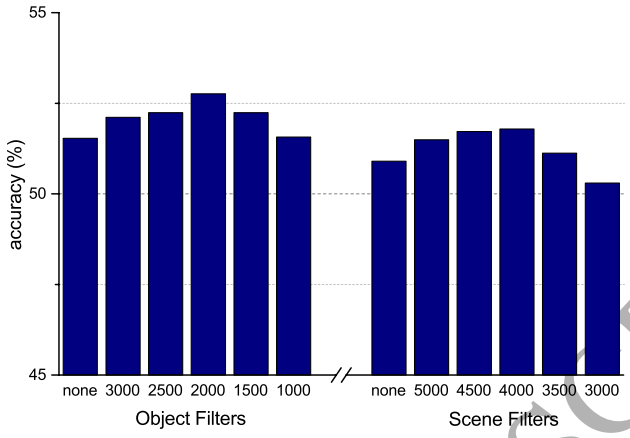


Figure 6: Classification accuracy on the MIT Indoor Scene dataset w.r.t. the number of clusters used in the clustering process. "none" indicates the direct use of all the original part filters. The original number of object filters and scene filters are 3,480 and 6,352 respectively. Clustering is able to remove 40% of the part filters with even better results.

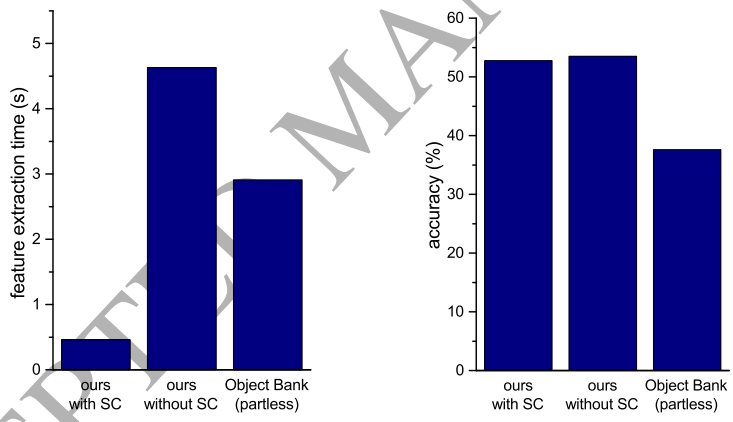


Figure 7: Effect of the sparse coding (SC) based acceleration. **Left.** Comparison of average feature extraction time per frame. **Right.** Comparison of classification accuracy on the MIT Indoor scene dataset. We adopt the fast implementation of Object Bank as an alternative method for comparison. By using this acceleration step, the feature extraction process is ten times faster and similar classification accuracy can be maintained.

ters. For the object part filters, compared with the situation without clustering ("none" in Figure 6), the accuracy increases first and then decreases after 2,000 clusters. This clearly validates the fact that there is redundant information in the original filters, and clustering is an effective way to alleviate the effect of

ACCEPTED MANUSCRIPT

this issue. When further reducing the number of clusters to less than 2,000, the performance drops as the final representation is not discriminative enough with too few filters. Very similar trend is observed on the scene filters. Generally around 40% of the filters can be safely reduced in this clustering process.

We also study the effect of sparse coding. Results are shown in Figure 7. We report the speed and accuracy of our representation using only the object filters, *i.e.*, "PMR (object only)". We also include the results of Object Bank as a comparison, which is also based on object detections but does not use parts (dubbed "partless"). Results show that our PMR representation with sparse coding based acceleration is ten times faster than the brute-force computation and six times faster than Object Bank. The performance degradation from applying this acceleration step is negligible.

5.5. Comparison with the State of the Arts

We compare our PMR with several state-of-the-art methods on all the four datasets. Results are summarized in Table 6 and Table 7. On Fifteen Scenes and UIUC Sports, our PMR representation outperforms most of the compared approaches, which cover a wide variety of techniques like matching kernel [46], feature encoding method [47, 48, 49] and saliency [50]. On MIT Indoor, it shows significantly better results than a recent method using mid-level concepts learned from Internet images [51]. Our approach outperforms [52], which selected training data with exemplar selection schemes and learned deep shared filter banks for scene recognition. Compared with [20, 21], where the authors proposed to construct mid-level representations using parts learned from the same dataset, our PMR is built on parts learned from external data. Although the performance of PMR is lower, it can be directly applied to new datasets without re-training like [20, 21]. On CCV, most of the recent state-of-the-art results [53, 54, 55] were obtained by fusing multi-modal features, *e.g.*, SIFT,

ACCEPTED MANUSCRIPT

STIP and audio descriptors. Our PMR offers competitive performance and fusing it with other representations like the audio clues is expected to further improve the results.

Table 6: Comparison with the state-of-the-art results on Fifteen Scenes and UIUC Sports. We also report the fusion results of our PMR (using both types of part filters) and the powerful CNN (VGG) features. Consistent improvements can be attained from fusion.

Fifteen Scenes		**UIUC Sports**	
Approach	**ACC**	**Approach**	**ACC**
Graph-matching [46]	82.1%	ScSPM [47]	82.7%
DDSFL [52]	84.4%	LSA [48]	82.3%
DSS [50]	85.5%	LPR [49]	86.3%
LPR [49]	85.8%	DDSFL [52]	86.9%
VGG [8]	**92.5%**	VGG [8]	**97.5%**
PMR	87.5%	PMR	88.1%
PMR+VGG	94.1%	PMR+VGG	97.7%

We also study the fusion performance of our PMR with the CNN feature. As reported in [7], the CNN feature achieved top results in a wide variety of visual recognition tasks, such as image classification, object detection, and visual instance retrieval. For instance, by using the "accurate" version of the *OverFeat* CNN model [58], which consists of 6 convolutional layers and 3 fully connected layers, Razavian *et al.* [7] achieved an accuracy of 69% on MIT Indoor.

In this work, instead of using the *OverFeat* model, we adopt a deeper and more powerful CNN architecture called *VGG-19* [8], which includes 16 convolutional layers and 3 fully connected layers. Note that although *VGG-19* is deeper, it has approximately the same number of parameters (144 millions) as *OverFeat*, because the convolutional kernels of *VGG-19* are smaller. We use the *Caffe* framework [59] along with the publicly available *Caffe* model of *VGG-19*[6]. The 4,096 dimensional outputs of a fully connected layer are used as the CNN representation.

[6]Model description by the authors of [8] is available at `https://gist.github.com/ksimonyan/3785162f95cd2d5fee77#file-readme-md`.

ACCEPTED MANUSCRIPT

Table 7: Comparison with the state-of-the-art results on MIT Indoor and CCV. We also report the fusion results of our PMR (using both types of part filters) and the powerful CNN (VGG) features. Consistent improvements can be attained from fusion.

MIT Indoor		**CCV**	
Approach	**ACC**	**Approach**	**mAP**
miSVM [51]	46.4%	Conceptlets [28]	62.5%
DDSFL [52]	52.3%	Jhuo *et al.* [53]	64.0%
IFV [20]	60.8%	HSVC [56]	67.3%
MLrep [21]	64.0%	Liu *et al.* [54]	68.2%
VGG [8]	77.8%	Wu *et al.* [55]	69.3%
Deep Filterbank [57]	**80.3%**	VGG [8]	**73.5%**
PMR	53.8%	PMR	60.1%
PMR+VGG	79.0%	PMR+VGG	75.4%

As shown in Table 6 and Table 7, the CNN features (VGG and Deep Filterbank) outperform all the other methods. This is not surprising as CNN has been proved to be very powerful in visual recognition tasks. By fusing our PMR with CNN, the performance is consistently improved on all the four datasets. Note that the fusion is done by averaging the prediction values from the two representations. Using advanced fusion weight learning method may further boost the performance. We also underline that, although CNN outperforms all the mid-level representations including PMR, mid-level features hold the advantage of being semantically interpretable. This enables advanced applications like text based visual search as we can match textual user queries to the semantics embedded in the representation.

6. Conclusions

In this paper, we have proposed a mid-level visual representation called PMR, which is based on the response maps of part filters. The proposed representation is generic and needs not to be fine-tuned on a certain target dataset. Through the extensive experiments on four benchmark image/video datasets, we first observed that our representation outperformed several representative

ACCEPTED MANUSCRIPT

low-level and mid-level image representations. Second, we observed that combining the representations from object part filters and scene part filters led to performance gains on large datasets like the CCV dataset. Third, we found that appropriate unsupervised clustering helped reduce the redundancy without losing useful information. By the virtue of sparse coding, we could accelerate feature extraction for ten times without hurting the recognition accuracy. Moreover, our representation could boost the performance of the state-of-the-art CNN feature using a simple fusion method. All these results corroborates that the proposed PMR is an effective and promising mid-level representation.

Acknowledgements

This work is supported in part by a National 863 Program (#2014AA015101) and a grant from the NSF China (#61572134).

References

References

[1] G. Csurka, C. Dance, L. Fan, J. Willamowski, C. Bray, Visual categorization with bags of keypoints, in: European Conference on Computer Vision Workshop on Statistical Learning in Computer Vision, no. 1-22, 2004, pp. 1–2.

[2] F. Perronnin, Y. Liu, J. Sánchez, H. Poirier, Large-scale image retrieval with compressed fisher vectors, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2010, pp. 3384–3391.

[3] D. G. Lowe, Distinctive image features from scale-invariant keypoints, International Journal of Computer Vision 60 (2) (2004) 91–110.

ACCEPTED MANUSCRIPT

[4] P. F. Felzenszwalb, R. B. Girshick, D. McAllester, D. Ramanan, Object detection with discriminatively trained part-based models, IEEE Transactions on Pattern Analysis and Machine Intelligence 32 (9) (2010) 1627–1645.

[5] M. Pandey, S. Lazebnik, Scene recognition and weakly supervised object localization with deformable part-based models, in: Proceedings of the IEEE International Conference on Computer Vision (ICCV), 2011, pp. 1307–1314.

[6] A. Krizhevsky, I. Sutskever, G. E. Hinton, Imagenet classification with deep convolutional neural networks, in: Advances in neural information processing systems (NIPS), 2012, pp. 1097–1105.

[7] A. Razavian, H. Azizpour, J. Sullivan, S. Carlsson, Cnn features off-the-shelf: an astounding baseline for recognition, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 2014, pp. 806–813.

[8] K. Simonyan, A. Zisserman, Very deep convolutional networks for large-scale image recognition, arXiv preprint arXiv:1409.1556.

[9] A. Farhadi, I. Endres, D. Hoiem, D. Forsyth, Describing objects by their attributes, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2009, pp. 1778–1785.

[10] A. Farhadi, I. Endres, D. Hoiem, Attribute-centric recognition for cross-category generalization, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2010, pp. 2352–2359.

[11] C. H. Lampert, H. Nickisch, S. Harmeling, Learning to detect unseen object classes by between-class attribute transfer, in: Proceedings of the IEEE

ACCEPTED MANUSCRIPT

Conference on Computer Vision and Pattern Recognition (CVPR), 2009, pp. 951–958.

[12] M. Rohrbach, M. Stark, G. Szarvas, I. Gurevych, B. Schiele, What helps where–and why? semantic relatedness for knowledge transfer, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2010, pp. 910–917.

[13] X. Yu, Y. Aloimonos, Attribute-based transfer learning for object categorization with zero/one training example, in: Proceedings of the European Conference on Computer Vision (ECCV), 2010, pp. 127–140.

[14] G. Wang, D. Forsyth, Joint learning of visual attributes, object classes and visual saliency, in: Proceedings of the IEEE International Conference on Computer Vision (ICCV), 2009, pp. 537–544.

[15] Y. Wang, G. Mori, A discriminative latent model of object classes and attributes, in: Proceedings of the European Conference on Computer Vision (ECCV), 2010, pp. 155–168.

[16] X. Kuang, N. Sang, C. Gao, Object localization with mid-level part detectors, Neurocomputing 212 (2016) 5864.

[17] J. Vogel, B. Schiele, Semantic modeling of natural scenes for content-based image retrieval, International Journal of Computer Vision 72 (2) (2007) 133–157.

[18] D. Parikh, K. Grauman, Relative attributes, in: Proceedings of the IEEE International Conference on Computer Vision (ICCV), 2011, pp. 503–510.

[19] S. Singh, A. Gupta, A. Efros, Unsupervised discovery of mid-level discriminative patches, in: Proceedings of the European Conference on Computer Vision (ECCV), 2012, pp. 73–86.

ACCEPTED MANUSCRIPT

[20] M. Juneja, A. Vedaldi, C. Jawahar, A. Zisserman, Blocks that shout: Distinctive parts for scene classification, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2013, pp. 923–930.

[21] C. Doersch, A. Gupta, A. A. Efros, Mid-level visual element discovery as discriminative mode seeking, in: Advances in neural information processing systems (NIPS), 2013, pp. 494–502.

[22] J. Sun, J. Ponce, Learning discriminative part detectors for image classification and cosegmentation, in: Proceedings of the IEEE International Conference on Computer Vision (ICCV), 2013, pp. 3400–3407.

[23] A. Hauptmann, R. Yan, W.-H. Lin, M. Christel, H. Wactlar, Can high-level concepts fill the semantic gap in video retrieval? a case study with broadcast news, IEEE Transactions on Multimedia 9 (5) (2007) 958–966.

[24] J. Liu, B. Kuipers, S. Savarese, Recognizing human actions by attributes, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2011, pp. 3337–3344.

[25] A. Diba, A. Mohammad Pazandeh, H. Pirsiavash, L. Van Gool, DeepCAMP: Deep Convolutional Action & Attribute Mid-Level Patterns, in: The IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016.

[26] F. Yuan, G.-S. Xia, H. Sahbi, V. Prinet, Mid-level features and spatio-temporal context for activity recognition, Pattern Recognition 45 (12) (2012) 4182–4191.

[27] W. Xu, Z. Miao, Y. Tian, A novel mid-level distinctive feature learning for action recognition via diffusion map, Neurocomputing 218 (2016) 185–196.

ACCEPTED MANUSCRIPT

ACCEPTED MANUSCRIPT

[28] M. Mazloom, E. Gavves, C. G. Snoek, Conceptlets: Selective semantics for classifying video events, IEEE Transactions on Multimedia 16 (8) (2014) 2214–2228.

[29] T. L. Berg, A. C. Berg, J. Shih, Automatic attribute discovery and characterization from noisy web data, in: Proceedings of the European Conference on Computer Vision (ECCV), 2010, pp. 663–676.

[30] L.-J. Li, H. Su, L. Fei-Fei, E. P. Xing, Object bank: A high-level image representation for scene classification & semantic feature sparsification, in: Advances in neural information processing systems (NIPS), 2010, pp. 1378–1386.

[31] L.-J. Li, H. Su, Y. Lim, L. Fei-Fei, Object bank: An object-level image representation for high-level visual recognition, International Journal of Computer Vision 107 (1) (2014) 20–39.

[32] Y. Zheng, Y.-G. Jiang, X. Xue, Learning hybrid part filters for scene recognition, in: Proceedings of the European Conference on Computer Vision (ECCV), 2012, pp. 172–185.

[33] N. Dalal, B. Triggs, Histograms of oriented gradients for human detection, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Vol. 1, 2005, pp. 886–893.

[34] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, L. Fei-Fei, Imagenet: A large-scale hierarchical image database, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2009, pp. 248–255.

[35] G. A. Miller, Wordnet: a lexical database for english, Communications of the ACM 38 (11) (1995) 39–41.

ACCEPTED MANUSCRIPT

[36] J. Xiao, J. Hays, K. A. Ehinger, A. Oliva, A. Torralba, Sun database: Large-scale scene recognition from abbey to zoo, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2010, pp. 3485–3492.

[37] J. Mairal, F. Bach, J. Ponce, G. Sapiro, Online dictionary learning for sparse coding, in: Proceedings of the International Conference on Machine Learning (ICML), 2009, pp. 689–696.

[38] J. Mairal, F. Bach, J. Ponce, G. Sapiro, Online learning for matrix factorization and sparse coding, The Journal of Machine Learning Research 11 (2010) 19–60.

[39] A. Quattoni, A. Torralba, Recognizing indoor scenes, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2009, pp. 413–420.

[40] S. Lazebnik, C. Schmid, J. Ponce, Beyond bags of features: Spatial pyramid matching for recognizing natural scene categories, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Vol. 2, 2006, pp. 2169–2178.

[41] L.-J. Li, L. Fei-Fei, What, where and who? classifying events by scene and object recognition, in: Proceedings of the IEEE International Conference on Computer Vision (ICCV), 2007, pp. 1–8.

[42] Y.-G. Jiang, G. Ye, S.-F. Chang, D. Ellis, A. C. Loui, Consumer video understanding: A benchmark database and an evaluation of human and machine performance, in: Proceedings of the ACM International Conference on Multimedia Retrieval (ICMR), 2011, p. 29.

ACCEPTED MANUSCRIPT

ACCEPTED MANUSCRIPT

[43] A. Oliva, A. Torralba, Modeling the shape of the scene: A holistic representation of the spatial envelope, International Journal of Computer Vision 42 (3) (2001) 145–175.

[44] L. Torresani, M. Szummer, A. Fitzgibbon, Efficient object category recognition using classemes, in: Proceedings of the European Conference on Computer Vision (ECCV), 2010, pp. 776–789.

[45] Y.-G. Jiang, Super: towards real-time event recognition in internet videos, in: Proceedings of the ACM International Conference on Multimedia Retrieval (ICMR), 2012, pp. 7:1–7:8.

[46] O. Duchenne, A. Joulin, J. Ponce, A graph-matching kernel for object categorization, in: Proceedings of the IEEE International Conference on Computer Vision (ICCV), 2011, pp. 1792–1799.

[47] J. Yang, K. Yu, Y. Gong, T. Huang, Linear spatial pyramid matching using sparse coding for image classification, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2009, pp. 1794–1801.

[48] L. Liu, L. Wang, X. Liu, In defense of soft-assignment coding, in: Proceedings of the IEEE International Conference on Computer Vision (ICCV), 2011, pp. 2486–2493.

[49] F. Sadeghi, M. F. Tappen, Latent pyramidal regions for recognizing scenes, in: Proceedings of the European Conference on Computer Vision (ECCV), 2012, pp. 228–241.

[50] G. Sharma, F. Jurie, C. Schmid, Discriminative spatial saliency for image classification, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2012, pp. 3506–3513.

ACCEPTED MANUSCRIPT

ACCEPTED MANUSCRIPT

[51] Q. Li, J. Wu, Z. Tu, Harvesting mid-level visual concepts from large-scale internet images, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2013, pp. 851–858.

[52] Z. Zuo, G. Wang, B. Shuai, L. Zhao, Q. Yang, Exemplar based deep discriminative and shareable feature learning for scene image classification, Pattern Recognition 48 (10) (2015) 3004–3015.

[53] I.-H. Jhuo, G. Ye, S. Gao, D. Liu, Y.-G. Jiang, D. Lee, S.-F. Chang, Discovering joint audio–visual codewords for video event detection, Machine Vision and Applications 25 (1) (2014) 33–47.

[54] D. Liu, K.-T. Lai, G. Ye, M.-S. Chen, S.-F. Chang, Sample-specific late fusion for visual category recognition, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2013, pp. 803–810.

[55] Z. Wu, Y.-G. Jiang, J. Wang, J. Pu, X. Xue, Exploring inter-feature and inter-class relationships with deep neural networks for video classification, in: Proceedings of the ACM International Conference on Multimedia (ACM MM), 2014, pp. 167–176.

[56] Y. Yang, H. Zhang, M. Zhang, F. Shen, X. Li, Visual coding in a semantic hierarchy, in: Proceedings of the ACM International Conference on Multimedia (ACM MM), 2015, pp. 59–68.

[57] M. Cimpoi, S. Maji, I. Kokkinos, A. Vedaldi, Deep filter banks for texture recognition, description, and segmentation, International Journal of Computer Vision (2015) 1–30.

[58] P. Sermanet, D. Eigen, X. Zhang, M. Mathieu, R. Fergus, Y. LeCun, Overfeat: Integrated recognition, localization and detection using convolu-

ACCEPTED MANUSCRIPT

tional networks, in: International Conference on Learning Representations (ICLR), 2014.

[59] Y. Jia, E. Shelhamer, J. Donahue, S. Karayev, J. Long, R. Girshick, S. Guadarrama, T. Darrell, Caffe: Convolutional architecture for fast feature embedding, in: Proceedings of the ACM International Conference on Multimedia (ACM MM), 2014, pp. 675–678.

ACCEPTED MANUSCRIPT

ACCEPTED MANUSCRIPT

Baodi Yuan is currently a Ph.D. candidate in School of Computer Science, Fudan University, Shanghai, China. His research interests include big video data analysis and data science

Jian Tu received the B.S. degree in Microelectronics in 2013 and M.S. degree in Computer Science in 2016, both from Fudan University, Shanghai, China. His research interests include computer vision and machine learning.

Rui-Wei Zhao is currently a Ph.D. candidate in School of Computer Science, Fudan University, Shanghai, China. He received his bachelor degree in automation in 2009 and master degree in control science in 2013, both from Tongji University, Shanghai, China. His current research interests include computer vision and machine learning.

ACCEPTED MANUSCRIPT

ACCEPTED MANUSCRIPT

Yingbin Zheng received the Ph.D. degree from Fudan University, Shanghai, China, in 2012. He is a Researcher of Shanghai Advanced Research Institue, Chinese Academy of Science, Shanghai, China. His current research interests include computer vision and multimedia information retrieval.

Yu-Gang Jiang is a Professor in School of Computer Science, Fudan University, Shanghai, China. He is interested in all aspects of extracting high-level information from big video data, such as video event recognition, object/scene recognition and large-scale visual search. He has received many awards, including ACM China Rising Star Award (2014) and ACM SIGMM Rising Star Award (2015).

ACCEPTED MANUSCRIPT